ACCENTURE 2026–2027 HIRING · ASSESSMENT 2

Technical Assessment Zero to Hero Guide

> 45 questions · 45 minutes · 4 sections · most elimination-heavy round_

Every topic in the syllabus explained from scratch: MS Office, Pseudocode and OOPs, Networking, Security and Cloud, and CS Fundamentals. Includes examples, tables, exam traps and 79 practice MCQs with explained answers.

A · MS Office

B · Pseudocode + OOPs

C · Networking · Security · Cloud

D · CS Fundamentals

45	45	79	100%
Questions	Minutes	Practice MCQs in this guide	Code answers verified by running them

Independent preparation guide · Not affiliated with or endorsed by Accenture

Edition: 2026–2027

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KNOW THE BATTLEFIELD

00 Exam Pattern & Strategy

Sub-section	≈ Qs	Topics	Difficulty
A · MS Office & Common Apps	7	Excel functions and formulas (VLOOKUP, IF, COUNTIF), Word formatting, PowerPoint features, output-based questions, shortcut keys	Easy marks
B · Pseudocode + OOPs	10	Bitwise operations, loops and nested loops, recursion tracing, string and array manipulation, OOPs concepts, 10–12 line code tracing with no pen and paper	Hardest
C · Networking, Security & Cloud	14	OSI and TCP/IP, IP addressing, HTTP/HTTPS/DNS/DHCP, firewalls and VPN, encryption, malware, IaaS/PaaS/SaaS, deployment models, CI/CD, Git, Docker	Concept recall
D · CS Fundamentals	14	Data types and operators, control structures, functions, arrays and strings, stack, queue and linked list, searching and sorting, time complexity, DBMS basics	Concept + logic

⏱ 1 minute per question

45 questions in 45 minutes. Recall-based questions (Parts A, C and D) should take about 30–40 seconds, which saves time for pseudocode.

🧠 No pen and paper

Practise tracing code in your head. Track only the variables that change, and work out the first 2–3 loop passes to spot the pattern.

🚫 Most eliminations

Candidates report this as the toughest filter. Parts C and D are pure memory, so revise them well: they're the easiest marks to secure.

Your target score

Section	Questions	Safe target	Strong target	Time budget
A · MS Office	7	5	7	~4 min (about 35 s each)
B · Pseudocode + OOPs	10	6	8+	~15 min (about 90 s each)
C · Networking, Security, Cloud	14	10	12+	~10 min (about 40 s each)
D · CS Fundamentals	14	10	12+	~12 min (about 50 s each)
Total	45	31 (≈ 70%)	39+ (≈ 87%)	~41 min, plus 4 min buffer

The official cut-off isn't published. These are self-set preparation targets, not official numbers.

Answering strategy

Read all 4 options before you answer. Eliminate the obviously wrong ones, because two options are often close. Don't get stuck: if a pseudocode question takes more than 2 minutes, make your best guess and move on. Negative marking hasn't been reported, but always confirm the rules shown on the test screen.

PART A · ≈ 7 QUESTIONS

MS Office & Common Applications

A

The fastest marks in the test. Most questions ask about an Excel formula's output, the purpose of a feature, or a shortcut key. Learn the patterns below and these become 20-second questions.

Excel formulas | VLOOKUP · IF · COUNTIF | Errors | Word | PowerPoint | Shortcuts

A1

EXCEL FOUNDATIONS

References, Formulas & Order of Operations

Formula basics. Every formula starts with `=`. A cell is named by column letter + row number (`B3`). A range uses a colon (`A1:C5`), and separate cells use commas (`SUM(A1,C1)`).

Cell references: the most-tested Excel concept

Type	Looks like	When copied to another cell...	Example: copy from C1 to D2
Relative	`A1`	Row and column both shift	`=A1` → `=B2`
Absolute	`$A$1`	Nothing shifts (it's locked)	`=$A$1` → `=$A$1`
Mixed (column locked)	`$A1`	Only the row shifts	`=$A1` → `=$A2`
Mixed (row locked)	`A$1`	Only the column shifts	`=A$1` → `=B$1`

Press `F4` while editing a reference to cycle through A1 → $A$1 → A$1 → $A1. **Rule:** the `$` sign "pins" whatever comes right after it.

Order of operations in Excel

Priority	Operator
1	`( )` brackets
2	`-` negation (unary minus)
3	`%` percent
4	`^` power
5	`*` `/`
6	`+` `-`
7	`&` join text
8	`= < > <= >= <>` comparison

Examples

`=2+3*4` → **14**
`=(2+3)*4` → **20**
`=10/2*5` → **25** (left to right)
`=2^3^2` → **64** (Excel works left to right: $(2^3)^2$)
`="5"&"5"` → **55** (text)

Trap: `=-2^2` gives **4** in Excel, not -4, because negation is applied *before* the power. Write `=-(2^2)` to get -4.

The sample sheet used in the examples and MCQs

	A	B	C
1	Name	Dept	Salary

	A	B	C
2	Asha	HR	40000
3	Ravi	IT	55000
4	Meena	IT	62000
5	John	Sales	38000
6	Priya	HR	45000

A1

MUST-KNOW FUNCTIONS

Excel Functions with Outputs

Function	What it does	Example (sample sheet) → result
`SUM` / `AVERAGE`	Adds / averages numbers	`=AVERAGE(C2:C6)` → 48000
`MAX` / `MIN`	Largest / smallest value	`=MAX(C2:C6)` → 62000
`LARGE` / `SMALL`	k-th largest / smallest	`=LARGE(C2:C6,2)` → 55000
`COUNT`	Counts **numeric** cells	`=COUNT(A1:C6)` → 5
`COUNTA`	Counts **non-empty** cells	`=COUNTA(A1:A6)` → 6
`COUNTBLANK`	Counts empty cells	—
`COUNTIF(range, criteria)`	Counts the cells that meet a condition	`=COUNTIF(C2:C6,">45000")` → 2
`COUNTIFS`	Counts with several conditions (AND)	`=COUNTIFS(B2:B6,"IT",C2:C6,">60000")` → 1
`SUMIF(range, criteria, sum_range)`	Adds values where a condition is true	`=SUMIF(B2:B6,"IT",C2:C6)` → 117000
`AVERAGEIF`	Averages values where a condition is true	`=AVERAGEIF(B2:B6,"HR",C2:C6)` → 42500
`IF(test, if_true, if_false)`	Returns one of two values	`=IF(C5>=40000,"OK","Low")` → Low
Nested `IF`	Several conditions	`=IF(C4>60000,"A",IF(C4>45000,"B","C"))` → A
`AND` / `OR` / `NOT`	Combine conditions (TRUE/FALSE)	`=AND(B3="IT",C3>50000)` → TRUE
`IFERROR(value, alt)`	Shows *alt* if there's an error	`=IFERROR(10/0,"NA")` → NA
`ROUND` / `ROUNDUP` / `ROUNDDOWN`	Round to n digits	`=ROUND(3.14159,2)` → 3.14 · `=ROUNDUP(2.01,0)` → 3
`INT` / `MOD`	Round down / remainder	`=INT(7.9)` → 7 · `=MOD(17,5)` → 2
`LEN`	Number of characters	`=LEN("Accenture")` → 9
`LEFT` / `RIGHT` / `MID`	Extract characters	`=MID("Accenture",3,4)` → "cent"
`UPPER` / `LOWER` / `PROPER`	Change case	`=PROPER("ravi kumar")` → Ravi Kumar

Function	What it does	Example (sample sheet) → result
`TRIM`	Removes extra spaces	`=TRIM(" a b ")` → "a b"
`CONCAT` / `&`	Joins text	`=A2&"-"&B2` → Asha-HR
`TODAY()` / `NOW()`	Current date / date + time	Recalculates automatically

Lookup functions

VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])
Searches the **first column** of the table and returns the value from column number *col_index_num* of the **same row**.
`FALSE` (or 0) = **exact match**. `TRUE` (or omitted) = **approximate match**, which needs the first column sorted ascending.
`=VLOOKUP("Ravi",A2:C6,2,FALSE)` → **IT** · `=VLOOKUP("Ravi",A2:C6,3,FALSE)` → **55000**

HLOOKUP
Same as VLOOKUP, but searches the first **row** and returns from row number n.

INDEX + MATCH
`=INDEX(C2:C6,MATCH("John",A2:A6,0))` → 38000. It can look left, which VLOOKUP cannot.

XLOOKUP (newer Excel)
`=XLOOKUP("John",A2:A6,C2:C6)` → 38000. Exact match by default.

VLOOKUP traps: (1) It only looks to the **right** of the first column. (2) A `col_index_num` larger than the table width gives `#REF!`. (3) A value that isn't found gives `#N/A`. (4) If you forget `FALSE`, it does an approximate match, which can return wrong results on unsorted data.

A2

RECOGNISE INSTANTLY

Excel Errors & Key Features

Error	Meaning	Typical cause
`#DIV/0!`	Division by zero	`=A1/0`, or dividing by an empty cell
`#N/A`	Value not available	VLOOKUP/MATCH didn't find the value
`#VALUE!`	Wrong data type	`="abc"+5`
`#REF!`	Invalid cell reference	A referenced cell was deleted, or the column index is out of range
`#NAME?`	Excel doesn't recognise the text	A misspelt function, like `=SUMM(A1:A3)`, or text without quotes
`#NUM!`	Invalid number	`=SQRT(-4)`
`#NULL!`	Ranges don't intersect	A space used instead of a comma between ranges
`####`	Not a real error	The column is too narrow to show the value

Pivot Table
Summarises large data by grouping (for example, total salary per department) without writing formulas.

Conditional Formatting
Changes how a cell looks (colour, icons, data bars) based on its value, for example highlighting salaries above 50000.

Data Validation
Restricts what can be typed in a cell: a drop-down list, number limits or dates.

Freeze Panes
Keeps header rows or columns visible while you scroll (View tab).

Filter & Sort
Filter *hides* rows that don't match. Sort *reorders* rows. Shortcut: `Ctrl+Shift+L`.

Charts
Use a **pie** chart for parts of a whole, a **line** chart for trends over time, a **column/bar** chart for comparisons and a **scatter** chart for relationships.

Flash Fill (`Ctrl+E`)
Spots a pattern from your typing and fills the rest of the column, for example extracting first names.

Goal Seek
Finds the input value needed to reach a target result (What-If Analysis).

Workbook vs worksheet
A workbook is the **file** (.xlsx). Worksheets are the **tabs** inside it.

BEYOND EXCEL

A3 Word, PowerPoint & Shortcuts

MS Word essentials

- **Styles** (Heading 1, 2…) give a consistent format and power the automatic **Table of Contents** (References tab).
- **Page break** starts a new page. A **section break** lets a part of the document have different headers, footers, orientation or margins.
- **Header/Footer** repeat on every page, for page numbers and titles.
- **Mail Merge** creates bulk personalised letters, labels or emails from a data list (Mailings tab).
- **Track Changes** and **Comments** are used for reviewing (Review tab).
- **Find & Replace** (`Ctrl+H`) and **Format Painter** (copies formatting).
- **Line spacing**, **indents** and **alignment**: left, centre, right, justify.

PowerPoint essentials

- **Slide Master** controls the theme, fonts, logo and layouts of **all** slides.
- **Animation** moves *objects* on a slide. **Transition** is the effect *between slides*.
- **Views:** Normal (editing), **Slide Sorter** (rearranging slides), Reading, Notes Page and Presenter View (speaker notes plus the next slide).
- **Speaker notes** are visible only to the presenter.
- **Hide Slide** keeps a slide in the file but skips it during the show.
- File types: `.pptx` (presentation), `.ppsx` (opens straight into the slideshow), `.potx` (template).

Shortcut	Action	Shortcut	Action
`Ctrl+Z / Ctrl+Y`	Undo / Redo	`F4` (Excel)	Toggle $ references
`Ctrl+C / X / V`	Copy / Cut / Paste	`F2` (Excel)	Edit active cell
`Ctrl+B / I / U`	Bold / Italic / Underline	`Alt+=` (Excel)	AutoSum

Shortcut	Action	Shortcut	Action
`Ctrl+F / Ctrl+H`	Find / Replace	`Ctrl+;` / `Ctrl+Shift+;`	Insert today's date / current time
`Ctrl+K`	Insert hyperlink	`Alt+Enter` (Excel)	New line inside a cell
`Ctrl+E / L / R / J` (Word)	Centre / Left / Right / Justify	`Ctrl+Space` / `Shift+Space`	Select column / row (Excel)
`Ctrl+Enter` (Word)	Page break	`Ctrl+D` / `Ctrl+R` (Excel)	Fill down / fill right
`F7`	Spelling & grammar check	`Ctrl+M` (PowerPoint)	New slide
`F12`	Save As	`F5` / `Shift+F5` (PPT)	Slideshow from start / from current slide
`Ctrl+P / Ctrl+S`	Print / Save	`Ctrl+D` (PowerPoint)	Duplicate slide

TEST YOURSELF

Part A · Practice MCQs

Questions A1–A7 use the sample sheet (Name / Dept / Salary in rows 2–6). Aim for under 40 seconds each.

QA1 Using the sample sheet above, what does `=COUNTIF(B2:B6,"IT")` return?

A 2
B 117000
C 1
D 3

QA2 Using the sample sheet above, what does `=SUMIF(B2:B6,"HR",C2:C6)` return?

A 2
B 40000
C 85000
D 45000

QA3 Using the sample sheet above, what does `=VLOOKUP("Meena",A2:C6,3,FALSE)` return?

A IT
B #N/A
C 62000
D Meena

QA4 Using the sample sheet above, what does `=AVERAGE(C2:C6)` return?

A 240000
B 45000
C 48000
D 50000

QA5 Using the sample sheet above, what does `=MAX(C2:C6)-MIN(C2:C6)` return?

A 17000
B 62000
C 24000
D 38000

QA6 Using the sample sheet above, what does `=IF(C3>50000,"High","Low")` return?

A Low
B 55000
C TRUE
D High

QA7 Using the sample sheet above, what does `=LEFT(A3,2)&RIGHT(A4,1)` return?

A Rav
B Raa
C Rae
D Ra

QA8 You copy the formula `=A1*$B$1` from C1 down to C3. What does C3 contain?

A =A3*$B$1
B =C3*$B$1
C =A1*$B$1
D =A3*$B$3

QA9 Which key toggles a cell reference between relative, absolute and mixed while you edit a formula?

A Ctrl + $
B F2
C F9
D F4

QA10 What does `=MID("Accenture",3,4)` return?

A Acce
B cent
C ente
D ccen

QA11 What does `=INT(-3.7)` return in Excel?

A -3.7
B -3
C 4
D -4

QA12 A cell shows #N/A after a VLOOKUP. What is the most likely reason?

A The lookup value was not found
B A deleted cell reference
C Division by zero
D The column is too narrow

QA13 Which is the difference between COUNT and COUNTA?

A They are identical
B COUNTA counts only blank cells
C COUNT counts text; COUNTA counts numbers
D COUNT counts numbers; COUNTA counts all non-empty cells

QA14 In MS Word, which shortcut inserts a page break?

A Ctrl + B
B Alt + Enter
C Shift + Enter
D Ctrl + Enter

QA15 Which Word feature creates many personalised letters from one template and a list of names?

A Mail Merge
B Track Changes
C Section Break
D Styles

QA16 In PowerPoint, which key starts the slideshow from the CURRENT slide?

A Alt + F5
B Ctrl + F5
C F5
D Shift + F5

QA17 Where do you change the font and logo for ALL slides in a presentation at once?

A Normal view
B Slide Master
C Reading view
D Slide Sorter

QA18 What is the difference between an Animation and a Transition in PowerPoint?

A Transition moves text; Animation changes slides
B They are the same
C Animation only works in Slide Sorter
D Animation moves objects on a slide; Transition is the effect between slides

QA19 Which Excel feature limits the entries in a cell to a drop-down list?

A Freeze Panes
B Data Validation
C Conditional Formatting
D Pivot Table

Part A · Answer key

Q	Ans	Correct option	Why
A1	**A**	2	COUNTIF counts the cells in B2:B6 equal to "IT": Ravi and Meena.
A2	**C**	85000	SUMIF adds the salaries in C where B is "HR": 40000 + 45000.
A3	**C**	62000	It finds "Meena" in the first column (exact match, because of FALSE) and returns the value from column 3 of the range.
A4	**C**	48000	(40000 + 55000 + 62000 + 38000 + 45000) ÷ 5 = 240000 ÷ 5.
A5	**C**	24000	62000 - 38000.
A6	**D**	High	C3 = 55000, which is more than 50000, so IF returns the 'value if true'.
A7	**B**	Raa	LEFT("Ravi", 2) = "Ra". RIGHT("Meena", 1) = "a". The & sign joins them into "Raa".
A8	**A**	=A3*$B$1	The relative reference A1 changes with the row (A3). The absolute reference $B$1 stays fixed.
A9	**D**	F4	F4 cycles A1 → $A$1 → A$1 → $A1.
A10	**B**	cent	MID starts at character 3 (the second c) and takes 4 characters: c-e-n-t.
A11	**D**	-4	INT rounds down to the nearest whole number. For negatives, 'down' means -4.
A12	**A**	The lookup value was not found	#N/A means 'not available': the value wasn't found. #DIV/0! is division by zero, #### means the column is too narrow, and #REF! is an invalid reference.
A13	**D**	COUNT counts numbers; COUNTA counts all non-empty cells	COUNT counts only numeric cells. COUNTA counts any non-empty cell, including text. COUNTBLANK counts empty cells.
A14	**D**	Ctrl + Enter	Ctrl + Enter adds a page break. Shift + Enter is a line break within a paragraph.
A15	**A**	Mail Merge	Mail Merge combines a main document with a data source, such as an Excel list.
A16	**D**	Shift + F5	F5 starts from the first slide. Shift + F5 starts from the current slide.
A17	**B**	Slide Master	Slide Master controls the layout and theme of every slide based on it.
A18	**D**	Animation moves objects on a slide; Transition is the effect between slides	Animations apply to objects such as text and images. Transitions apply when you move from one slide to the next.
A19	**B**	Data Validation	Data → Data Validation → List restricts input. Conditional Formatting only changes how cells look.

PART B · ≈ 10 QUESTIONS · HARDEST SECTION

Pseudocode + OOPs

B

Short programs, 10–12 lines long, that you trace in your head and then predict the output. They test bitwise logic, loops, recursion and arrays, plus OOPs concepts. This section is where most marks are won or lost.

Operators | Bitwise & | ^ ~ << >> | Loops | Recursion | Strings & arrays | OOPs

B1

LEARN THE LANGUAGE

Reading Pseudocode & Operators

Pseudocode is language-neutral, but it behaves like C or Java. These are the conventions you'll see:

```
// declarations
Integer a = 5, b
Set b = a + 2          // assignment
if (a > b) ... else ... end if
for (i = 1 to 5) ... end for  // 1..5 inclusive
while (cond) ... end while
Function f(Integer n) ... return x
Print a, b
```

Operator	Meaning
`/` on integers	Integer division: `7/2 = 3`
`mod` or `%`	Remainder: `17 mod 5 = 2`
`&&` `||` `!`	Logical AND, OR, NOT
`&` `|` `^` `~`	Bitwise AND, OR, XOR, NOT
`<<` `>>`	Left / right shift
`==` vs `=`	Comparison vs assignment

Operator precedence (C/Java style, highest first)

Level	Operators	Example
1	`()` `[]`	—
2	Unary `!` `~` `++` `--` `-`	`~5 = -6`
3	`*` `/` `%`	`2 + 3 * 4 = 14`
4	`+` `-`	`10 - 4 - 3 = 3` (left to right)
5	`<<` `>>`	`1 << 2 + 1 = 1 << 3 = 8`
6	`<` `<=` `>` `>=` then `==` `!=`	—
7	`&` then `^` then `|`	`5 | 2 & 3 = 5 | 2 = 7`
8	`&&` then `||`	—
9	`?:`, then assignment `=` `+=` `-=`	—

Two classic traps: (1) `+` binds tighter than `<<`, so `1 << 2 + 1` is 8, not 5. (2) `==` binds tighter than `&`, so `x & 1 == 0` means `x & (1 == 0)`. When unsure, assume the question wants the precedence rules applied literally.

Increment operators. `x++` uses the value **first**, then adds 1. `++x` adds 1 **first**. With `a = 5`: `b = a++` gives b = 5, a = 6, and `b = ++a` gives b = 6, a = 6.

B2

THE #1 TOPIC

Binary & Bitwise Operations

Decimal → binary in seconds

Write the place values **16 8 4 2 1** and mark the ones that add up to your number.

n	16	8	4	2	1	Binary
5	0	0	1	0	1	`00101`
12	0	1	1	0	0	`01100`
13	0	1	1	0	1	`01101`
25	1	1	0	0	1	`11001`

Truth table

a	b	a&b	a|b	a^b
0	0	0	0	0
0	1	0	1	1
1	0	0	1	1
1	1	1	1	0

AND = both 1 · OR = at least one 1 · XOR = **different** bits

Worked examples with a = 12 (1100) and b = 10 (1010)

Expression	Binary working	Result
`a & b`	1100 & 1010 = 1000	**8**
`a | b`	1100 | 1010 = 1110	**14**
`a ^ b`	1100 ^ 1010 = 0110	**6**
`~a`	−(a + 1)	**−13**
`a << 2`	110000 = 12 × 2^2	**48**
`a >> 2`	11 = 12 ÷ 2^2	**3**

Shortcuts that save minutes

Shifts
`x << n = x × 2ⁿ`
`x >> n = x ÷ 2ⁿ` (integer)

NOT
`~x = −(x + 1)`
`~0 = −1` , `~(−1) = 0`

XOR rules
`x ^ x = 0` · `x ^ 0 = x`
Three XORs swap two variables

Odd or even
`x & 1` = 1 if odd, 0 if even

Power of 2?
`x & (x − 1) == 0` for 1, 2, 4, 8...

Identity
`a + b = (a ^ b) + 2·(a & b)`
`(a | b) = (a ^ b) + (a & b)`

Remember: `x & (x − 1)` clears the lowest set bit. A loop that does this until x becomes 0 **counts the 1-bits**. So does a loop of `c += x & 1; x >>= 1`.

B3

TRACE WITHOUT PAPER

Loops, Recursion, Arrays & Strings

The trace-table technique

```
Integer s = 0, i = 1
while (i <= 10)
    s = s + i
    i = i * 2
end while
Print s
```

Pass	i (start)	s after	i after
1	1	1	2
2	2	3	4
3	4	7	8
4	8	15	16 → stop

Output: **15**. In your head, keep a mini table of only the variables that change. After 2–3 passes, look for the pattern (here i doubles, so s = 1 + 2 + 4 + 8).

Loop-count formulas

Loop	Iterations
`for i = 1 to n`	n
`for i = a to b`	b − a + 1
`for i = 1 to n`, then inside `for j = 1 to n`	n^2
`for i = 1 to n`, then inside `for j = i to n`	n(n + 1)/2
`i = 1; while (i < n) i = i * 2`	$\lceil \log_2 n \rceil$
`break` / `continue`	break leaves the innermost loop; continue skips to the next pass

Recursion: unwind it on the way down, add it up on the way back

```
Function f(Integer n)
    if (n == 0) return 0
    return n mod 10 + f(n / 10)
Print f(472)
```

Trace (digit sum)

f(472) = 2 + f(47)
f(47) = 7 + f(4)
f(4) = 4 + f(0) = 4 + 0
Back up: 4 → 11 → **13**

Factorial

`f(n) = n × f(n−1)`, f(0) = 1 → f(5) = 120

Fibonacci

`f(n) = f(n−1) + f(n−2)` → 0 1 1 2 3 5 8 13…

GCD (Euclid)

`g(a, b) = g(b, a mod b)`, g(a, 0) = a

Print order trap: if a function prints *before* its recursive call, the output comes in forward order. If it prints *after* the call, the output comes in **reverse** order. For example, printing n after `f(n−1)` with n = 3 gives "1 2 3".

Arrays & strings

- Indexes usually start at **0**: an array of 5 elements runs from index 0 to 4. Going past the end is an **out-of-bounds** error.
- Common patterns: running max/min, swapping `a[i]` and `a[n−1−i]` to reverse, counting characters, checking palindromes.

- For a string s, `length(s)` gives the number of characters. `s[i]` is the i-th character, and `'A'` has ASCII code 65 (`'a'` = 97, `'0'` = 48).

OBJECT-ORIENTED PROGRAMMING

B4 OOPs: The Four Pillars & Beyond

Class vs Object. A **class** is a blueprint (for example, `Car`). An **object** is a real instance created from it (`Car c = new Car()`). The class defines **attributes** (data) and **methods** (behaviour).

Pillar	Meaning	Example
Encapsulation	Bundles data and methods together and hides internal state	`private int balance;` accessed only through `getBalance()` / `deposit()`
Abstraction	Shows *what* something does and hides *how*	`abstract class Shape { abstract double area(); }`, or interfaces
Inheritance	A child class reuses a parent's members (an "is-a" relationship)	`class Dog extends Animal`
Polymorphism	One name, many forms	Overloading (compile time) and overriding (run time)

Overloading (compile time)

Same method name in the **same class**, with different **parameters**.

`add(int a, int b)` and `add(double a, double b)`

Overriding (run time)

A child **redefines** a parent method with the **same signature**. Which version runs depends on the **actual object**:

`Animal a = new Dog(); a.sound();` runs Dog's sound().

Inheritance type	Shape
Single	A → B
Multilevel	A → B → C
Hierarchical	A → B and A → C
Multiple	B, C → D (in Java, only through **interfaces**)
Hybrid	A combination of the above

Access modifier	Visible in
`private`	Same class only
default (none)	Same package
`protected`	Same package and subclasses
`public`	Everywhere

Constructor

Same name as the class, **no return type**, runs automatically when an object is created. It can be overloaded.

Abstract class vs Interface

An abstract class can have state and concrete methods. A class can **extend one** abstract class but **implement many** interfaces.

static / this / super

`static` belongs to the class, not to objects. `this` is the current object. `super` refers to the parent's members or constructor.

Also know: `final` class = can't be inherited · `final` method = can't be overridden · **association** (uses-a), **aggregation** (has-a, parts can exist alone) and **composition** (has-a, parts die with the whole).

TEST YOURSELF

Part B · Practice MCQs

B1–B18 are pseudocode questions. **Each answer was verified by actually running the code.** B19–B26 cover OOPs. Trace in your head first, and aim for under 90 seconds each.

QB1 Bitwise XOR and AND — What is the output?

```
Integer a = 5, b = 9
a = a ^ b
b = a & b
Print a + b
```

A 20
B 13
C 14
D 17

QB2 Counting set bits — What is the output?

```
Integer x = 13, count = 0
while (x > 0)
    count = count + (x & 1)
    x = x >> 1
end while
Print count
```

A 3
B 13
C 4
D 2

QB3 Nested loop count — What is the output?

```
Integer s = 0
for (i = 1 to 4)
    for (j = i to 4)
        s = s + 1
    end for
end for
Print s
```

A 8
B 16
C 6
D 10

QB4 Recursion — What is the output?

```
Function fun(Integer n)
    if (n <= 1) return 1
    return n + fun(n - 2)
End Function
Print fun(7)
```

A 15
B 13
C 28
D 16

QB5 mod and integer division — What is the output?

```
Integer a = 10, b = 3
Integer c = a mod b + a / b
Print c
```

A 4
B 1
C 3
D 4.33

QB6 Shift operators — What is the output?

```
Integer p = 6
p = (p << 2) + (p >> 1)
Print p
```

A 30
B 26
C 27
D 15

QB7 Digit reversal — What is the output?

```
Integer n = 1234, r = 0
while (n != 0)
    r = r * 10 + n mod 10
    n = n / 10
end while
Print r
```

A 4123
B 1234
C 4321
D 10

QB8 Bitwise NOT (~) — What is the output?

```
Integer a = 7
Print ~a + (a & 3)
```

A 11
B -4
C -8
D -5

QB9 Array traversal — What is the output?

```
Integer arr[5] = {3, 8, 1, 9, 4}
Integer m = arr[0]
for (i = 1 to 4)
    if (arr[i] > m) m = arr[i]
    else m = m - 1
end for
Print m
```

A 8
B 3
C 9
D 7

QB10 Recursion with mod — What is the output?

```
Function f(Integer n)
    if (n == 0) return 0
    return (n mod 2) + f(n / 2)
End Function
Print f(29)
```

A 4
B 3
C 29
D 5

QB11 Loop with two variables — What is the output?

```
Integer x = 2, y = 5
for (i = 1 to 3)
    x = x * y
    y = y + x mod 7
end for
Print x, y
```

A 800 16
B 880 16
C 880 11
D 80 11

QB12 Euclid's GCD — What is the output?

```
Function g(Integer a, Integer b)
    if (b == 0) return a
    return g(b, a mod b)
End Function
Print g(84, 36)
```

A 6
B 36
C 12
D 4

QB13 String indexing — What is the output?

```
String s = "ACCENTURE"
Integer c = 0
for (i = 0 to length(s) - 1)
    if (s[i] == 'E') c = c + i
end for
Print c
```

A 8
B 2
C 12
D 11

QB14 XOR swap — What is the output?

```
Integer a = 12, b = 25
a = a ^ b
b = a ^ b
a = a ^ b
Print a - b
```

A -13
B 0
C 13
D 37

QB15 Nested loop with break — What is the output?

```
Integer s = 0
for (i = 1 to 5)
    for (j = 1 to 5)
        if (j > i) break
        s = s + j
    end for
end for
Print s
```

A 25
B 15
C 35
D 55

QB16 Fast power — What is the output?

```
Function p(Integer b, Integer e)
    if (e == 0) return 1
    if (e mod 2 == 0) return p(b, e/2) * p(b, e/2)
    return b * p(b, e - 1)
End Function
Print p(3, 5)
```

A 81
B 125
C 15
D 243

QB17 AND, OR, XOR together — What is the output?

```
Integer a = 10, b = 6
Print (a & b) + (a | b) - (a ^ b)
```

A 16
B 4
C 2
D 14

QB18 While loop with if/else — What is the output?

```
Integer n = 100, steps = 0
while (n > 1)
    if (n mod 2 == 0) n = n / 2
    else n = 3 * n + 1
    steps = steps + 1
end while
Print steps
```

A 24
B 7
C 100
D 25

QB19 Two methods in the same class have the same name but different parameter lists. This is called:

A Method overloading
B Method overriding
C Encapsulation
D Abstraction

QB20 Run-time (dynamic) polymorphism is achieved through:

A Method overriding
B Constructors
C Method overloading
D Static methods

QB21 Making fields private and giving access through getters and setters is an example of:

A Abstraction
B Encapsulation
C Polymorphism
D Inheritance

QB22 In Java, a class cannot extend two classes. Multiple inheritance of type is achieved using:

A Packages
B Final classes
C Abstract classes
D Interfaces

QB23 What is printed?

```
class A { void show() { print("A") } }
class B extends A { void show() { print("B") } }
A obj = new B()
obj.show()
```

A B
B AB
C Compile error
D A

QB24 Which of these cannot be instantiated (you can't create an object of it directly)?

A A final class
B An abstract class
C A class with a constructor
D A subclass

QB25 Which statement about constructors is TRUE?

A It must return void
B It has the same name as the class and no return type
C It can be called only once per program
D It must be static

QB26 **Which access modifier makes a member visible ONLY inside its own class?**

A protected
B private
C public
D default (package)

Part B · Answer key

Q	Ans	Correct option	Why
B1	**A**	20	a = 0101 ^ 1001 = 1100 = 12. b = 1100 & 1001 = 1000 = 8. 12 + 8 = 20.
B2	**A**	3	13 = 1101 in binary. The loop adds the last bit each time and shifts right, so it counts the 1s: 3.
B3	**D**	10	The inner loop runs 4, 3, 2, 1 times: 4 + 3 + 2 + 1 = 10.
B4	**D**	16	fun(7) = 7 + fun(5) = 7 + 5 + fun(3) = 7 + 5 + 3 + fun(1) = 7 + 5 + 3 + 1 = 16.
B5	**A**	4	10 mod 3 = 1. 10 / 3 = 3 (integer division drops the decimal). 1 + 3 = 4.
B6	**C**	27	6 << 2 = 6 × 4 = 24. 6 >> 1 = 6 ÷ 2 = 3. 24 + 3 = 27.
B7	**C**	4321	Each step takes the last digit (n mod 10) and appends it to r, so the number is reversed: 4321.
B8	**D**	-5	~a = -(a + 1) = -8 (two's complement). 7 & 3 = 0111 & 0011 = 3. -8 + 3 = -5.
B9	**A**	8	m = 3 → 8 (8 > 3) → 7 (1 is not bigger, so subtract 1) → 9 (9 > 7) → 8 (4 is not bigger). Answer: 8.
B10	**A**	4	This adds the binary digits of n. 29 = 11101, which has four 1s, so the answer is 4.
B11	**B**	880 16	Pass 1: x = 10, y = 5 + 3 = 8. Pass 2: x = 80, y = 8 + 3 = 11. Pass 3: x = 880, y = 11 + (880 mod 7 = 5) = 16.
B12	**C**	12	g(84, 36) → g(36, 12) → g(12, 0) → 12. This is Euclid's algorithm for the greatest common divisor.
B13	**D**	11	Indexes start at 0. The two E's are at positions 3 and 8, and 3 + 8 = 11.
B14	**C**	13	Three XORs swap the values, so a = 25 and b = 12. 25 - 12 = 13.
B15	**C**	35	For each i, the inner loop adds 1 + 2 + ... + i. So the total is 1 + 3 + 6 + 10 + 15 = 35.
B16	**D**	243	p(3, 5) works out 3^5 using squaring: 3 × p(3, 4) = 3 × $(p(3, 2))^2$ = 3 × 9^2 = 243.
B17	**B**	4	1010 & 0110 = 0010 = 2. 1010 | 0110 = 1110 = 14. 1010 ^ 0110 = 1100 = 12. 2 + 14 - 12 = 4.
B18	**D**	25	This is the Collatz sequence. Trace it: 100 → 50 → 25 → 76 → 38 → 19 → ... → 1, which takes 25 steps.
B19	**A**	Method overloading	Overloading is compile-time polymorphism within one class.
B20	**A**	Method overriding	The overridden method that runs is chosen at run time, based on the actual object.

Q	Ans	Correct option	Why
B21	**B**	Encapsulation	Encapsulation means bundling data with its methods and hiding the internal state.
B22	**D**	Interfaces	A class can implement several interfaces.
B23	**A**	B	The object is really a B, so B's overriding show() runs. This is run-time polymorphism.
B24	**B**	An abstract class	Abstract classes can only be extended. Their subclasses can be instantiated.
B25	**B**	It has the same name as the class and no return type	Constructors initialise objects. They have no return type, not even void.
B26	**B**	private	The scope widens in this order: private < default < protected < public.

PART C · ≈ 14 QUESTIONS

Networking, Security & Cloud

C

Mostly recall: layers, ports, protocols, attack types, cloud models and DevOps tools. With the tables in this part memorised, this becomes the highest-scoring section.

OSI & TCP/IP | IP & subnetting | HTTP · DNS · DHCP | Security | IaaS · PaaS · SaaS | CI/CD · Git · Docker

C1

THE REFERENCE MODELS

OSI & TCP/IP Models, Devices, Ports

#	OSI layer	Job	Protocols / examples	Device	Data unit
7	**Application**	Network services for the user	HTTP, HTTPS, FTP, SMTP, DNS, DHCP	—	Data
6	**Presentation**	Format, encrypt, compress	SSL/TLS, JPEG, ASCII	—	Data
5	**Session**	Open, manage and close sessions	NetBIOS, RPC	—	Data
4	**Transport**	End-to-end delivery, ports, reliability	TCP, UDP	—	Segment / datagram
3	**Network**	Logical addressing and **routing**	IP, ICMP (ping)	Router	Packet
2	**Data Link**	MAC addressing, framing, error detection	Ethernet, PPP, ARP*	Switch, bridge	Frame
1	**Physical**	Bits over the medium	Cables, Wi-Fi radio	Hub, repeater	Bits

*ARP maps IP → MAC and sits between Layers 2 and 3. Memory trick, top to bottom: "**A**ll **P**eople **S**eem **T**o **N**eed **D**ata **P**rocessing".

TCP/IP layer	= OSI layers
Application	7, 6, 5
Transport	4
Internet	3
Network Access (Link)	2, 1

Port	Protocol	Port	Protocol
20/21	FTP	80	HTTP
22	SSH (secure remote login)	110	POP3
23	Telnet (insecure)	143	IMAP
25	SMTP (send email)	443	HTTPS
53	DNS	3306	MySQL
67/68	DHCP	3389	RDP

Hub vs Switch vs Router
A hub sends to **all** ports (L1). A switch sends to the **right device** using the MAC address (L2). A router connects **different networks** using IP (L3).

MAC vs IP
A MAC address is a 48-bit **physical** address burnt into the NIC. An IP address is a **logical** address that can change. ARP finds the MAC for an IP.

LAN / MAN / WAN
Building → city → country or world. The Internet is the largest WAN. Topologies: bus, star (most common), ring, mesh, tree.

C2

ADDRESSES

IP Addressing & Subnetting

IPv4 = 32 bits, written as 4 octets (`192.168.1.10`), each 0–255. **IPv6** = 128 bits, written in hexadecimal (`2001:db8::1`), created because IPv4 addresses ran out.

Class	First octet	Default mask
A	1 – 126	255.0.0.0 (/8)
B	128 – 191	255.255.0.0 (/16)
C	192 – 223	255.255.255.0 (/24)
D	224 – 239	Multicast
E	240 – 255	Experimental

Special range	Purpose
10.0.0.0/8	Private
172.16.0.0 – 172.31.255.255	Private
192.168.0.0/16	Private (home routers)
127.0.0.1	Loopback (localhost)
169.254.x.x	APIPA (DHCP failed)

Subnetting in 3 formulas

Host bits
h = 32 − prefix
/26 → h = 6

Usable hosts
$2^h - 2$
/26 → 64 − 2 = **62**

Block size
256 − last mask octet
/26 = .192 → blocks of **64**

Prefix	Mask	Addresses	Usable hosts
/24	255.255.255.0	256	254
/25	255.255.255.128	128	126
/26	255.255.255.192	64	62
/27	255.255.255.224	32	30
/28	255.255.255.240	16	14
/30	255.255.255.252	4	2 (point-to-point links)

Two addresses in every subnet are reserved: the **network address** (all host bits 0) and the **broadcast address** (all host bits 1). Example: in 192.168.1.64/26, the network is .64, the usable hosts are .65–.126 and the broadcast is .127.

C3

HOW THE WEB WORKS

HTTP, DNS, DHCP & TCP vs UDP

	TCP	UDP
Connection	Connection-oriented (3-way handshake: **SYN → SYN-ACK → ACK**)	Connectionless
Reliability	Guaranteed, ordered, retransmits lost data	No guarantee, no ordering
Speed	Slower (more overhead)	Faster (less overhead)
Used for	Web (HTTP/S), email, file transfer, SSH	Streaming, gaming, VoIP, DNS queries

HTTP

- **HTTP** sends plain text on port 80. **HTTPS** = HTTP + TLS encryption on port 443 (the padlock icon).
- **Methods:** GET (read), POST (create), PUT (replace), PATCH (modify), DELETE.
- **Status codes:** 1xx info · **2xx success** (200 OK, 201 Created) · **3xx redirect** (301 moved permanently) · **4xx client error** (400 bad request, 401 unauthorised, 403 forbidden, 404 not found) · **5xx server error** (500 internal error, 503 unavailable).
- HTTP is **stateless**. Cookies and sessions remember users between requests.

DNS & DHCP

- **DNS** translates names to IP addresses (www.example.com → 93.184.x.x). It checks the browser cache, then the OS, then a recursive resolver, then the root, TLD and authoritative servers.
- **DNS records:** A (IPv4), AAAA (IPv6), CNAME (alias), MX (mail server), NS (name server), TXT.
- **DHCP** automatically gives a device its IP address, subnet mask, gateway and DNS using **DORA**: Discover → Offer → Request → Acknowledge.
- **ICMP** is used by `ping` and `traceroute`. **NAT** maps private IPs to one public IP.

C4

PROTECT THE SYSTEM

Security: Encryption, Malware, Attacks

CIA triad

Confidentiality: only authorised people can see the data.
Integrity: the data isn't altered.
Availability: the system is accessible when needed.

AuthN vs AuthZ

Authentication answers "who are you?" (password, OTP, biometrics). **Authorization** answers "what are you allowed to do?" (permissions). **MFA** combines 2 or more factors.

Firewall & VPN

A **firewall** filters traffic by rules (packet-filtering, stateful, application/WAF). A **VPN** creates an encrypted tunnel over the internet. A **proxy** forwards requests on your behalf.

Technique	How it works	Examples
Symmetric encryption	The **same key** encrypts and decrypts. It's fast, but the key must be shared securely.	AES, DES, 3DES
Asymmetric encryption	A **public key** encrypts and the **private key** decrypts. It solves key sharing but is slower.	RSA, ECC, Diffie–Hellman (key exchange)
Hashing	A **one-way** fixed-length fingerprint that can't be reversed. Used to store passwords and check integrity.	SHA-256, (MD5: weak)
Digital signature	The sender hashes the data and signs it with their **private** key. Anyone can verify it with the public key. It proves authenticity and integrity.	TLS certificates

Malware	Key trait
Virus	Attaches to a host file and spreads when it runs
Worm	Self-replicates over networks, **no host needed**
Trojan	Disguised as legitimate software
Ransomware	Encrypts files and demands payment
Spyware / Keylogger	Secretly records activity or keystrokes
Adware	Unwanted ads

Attack	What happens
Phishing	Fake email or site tricks you into giving data
DoS / DDoS	Floods a service so real users can't reach it
Man-in-the-Middle	Intercepts communication between two parties
SQL Injection	Malicious SQL in an input field (`' OR 1=1`)
XSS	Injects scripts into web pages viewed by others

Malware	Key trait
Rootkit	Hides deep in the OS to keep admin control
Botnet	A network of infected devices controlled remotely

Attack	What happens
Brute force	Tries many passwords until one works
Social engineering	Manipulates people, not systems
Zero-day	Exploits a flaw before a fix exists

MODERN INFRASTRUCTURE

C5 Cloud & DevOps

Cloud computing means on-demand IT resources over the internet with pay-as-you-go pricing. NIST's **5 characteristics**: on-demand self-service, broad network access, resource pooling, rapid elasticity and measured service.

Model	You manage	Provider manages	Examples
IaaS	OS, runtime, apps, data	Hardware, network, virtualisation	AWS EC2, Azure VMs, Google Compute Engine
PaaS	Apps and data only	Everything up to the runtime	Google App Engine, Heroku, Azure App Service
SaaS	Nothing (you just use it)	Everything	Gmail, Microsoft 365, Salesforce, Zoom
FaaS (serverless)	Function code	Servers, scaling	AWS Lambda, Azure Functions

Deployment model	Meaning
Public	Shared provider infrastructure (AWS, Azure, GCP)
Private	Dedicated to one organisation
Hybrid	Private + public combined
Community	Shared by organisations with common needs
Multi-cloud	Using several public providers

Virtualisation
A **hypervisor** runs several VMs on one physical server.

VM vs Container
A VM includes a full guest OS (heavy). A container shares the host **kernel** (light and fast to start).

Scaling
Vertical = a bigger machine. **Horizontal** = more machines. **Elasticity** = scaling automatically with demand.

DevOps, CI/CD, Git & Docker

DevOps brings development and operations together to deliver faster and more reliably. Pipeline: **Plan → Code → Build → Test → Release → Deploy → Operate → Monitor**. **CI** (Continuous Integration) means merging code often and automatically building and testing each change. **CD** means Continuous Delivery (always ready to release) or Continuous Deployment (released automatically). Tools: Jenkins, GitHub Actions, GitLab CI.

Git command	Does
`git init`	Create a new local repo

Docker term	Meaning
Dockerfile	Recipe of steps to build an image

Git command	Does
`git clone <url>`	Copy a remote repo locally
`git status`	Show changed and staged files
`git add`	Stage changes
`git commit -m`	Save a snapshot locally
`git push` / `git pull`	Upload / download + merge
`git fetch`	Download without merging
`git branch` / `git checkout`	Create / switch branches
`git merge` / `git stash`	Combine branches / shelve changes

Docker term	Meaning
Image	Read-only template (app + dependencies)
Container	A **running instance** of an image
Registry	Stores images (Docker Hub)
`docker build / run`	Build an image / start a container
`docker ps`	List running containers
Kubernetes	Orchestrates many containers (scaling, healing)

SDLC & Agile. **Waterfall** runs in sequential phases. **Agile** works in short iterations. **Scrum** uses sprints of 1–4 weeks, a Product Owner, a Scrum Master, daily stand-ups and a backlog.

TEST YOURSELF

Part C · Practice MCQs

These are pure recall questions. Aim for under 30 seconds each, and revise the tables above for any you miss.

QC1 **Which OSI layer is responsible for routing packets between networks?**

A Data Link (Layer 2)
B Network (Layer 3)
C Transport (Layer 4)
D Session (Layer 5)

QC2 **A switch mainly operates at which OSI layer?**

A Application (Layer 7)
B Data Link (Layer 2)
C Network (Layer 3)
D Physical (Layer 1)

QC3 **Default port number for HTTPS:**

A 21
B 8080
C 443
D 80

QC4 **Which protocol automatically assigns IP addresses to devices?**

A FTP
B DNS
C ARP
D DHCP

QC5 **Which is a PRIVATE IPv4 address?**

A 192.168.1.10
B 11.0.0.1
C 8.8.8.8
D 172.32.0.5

QC6 **How many usable host addresses are in a /26 subnet?**

A 64
B 62
C 30
D 126

QC7 **Which statement about TCP and UDP is correct?**

A TCP is connection-oriented and reliable; UDP is connectionless and faster
B UDP guarantees delivery; TCP does not
C Both are Network-layer protocols
D TCP is used for DNS queries by default; UDP for web pages

QC8 **HTTP status code 404 means:**

A Internal Server Error
B Not Found
C Forbidden
D OK

QC9 **Which is an example of SYMMETRIC encryption?**

A SHA-256
B RSA
C AES
D Diffie–Hellman

QC10 **Malware that encrypts your files and demands payment is:**

A Spyware
B Worm
C Adware
D Ransomware

QC11 **An attacker inserts `' OR 1=1 --` into a login form. This attack is:**

A Phishing
B DDoS
C SQL Injection
D Cross-Site Scripting

QC12 **The 'CIA triad' in security stands for:**

A Confidentiality, Identity, Authentication
B Confidentiality, Integrity, Availability
C Control, Identity, Access
D Cyber, Internet, Access

QC13 **Renting virtual machines from AWS EC2 is an example of:**

A FaaS
B SaaS
C PaaS
D IaaS

QC14 **Using Gmail or Microsoft 365 in a browser is an example of:**

A On-premise
B SaaS
C PaaS
D IaaS

QC15 **A company keeps sensitive data in its private cloud and uses a public cloud for extra capacity. This is:**

A Community cloud
B Hybrid cloud
C Public cloud
D Multi-tenant SaaS

QC16 **Which Git command creates a local copy of a remote repository?**

A git commit
B git clone
C git pull
D git init

QC17 **In Docker, what is a container?**

A A file that lists build steps
B A virtual machine with its own full OS kernel
C A cloud provider
D A running instance of an image

QC18 **Continuous Integration (CI) mainly means:**

A Merging code often, with automated build and tests on each change
B Manually testing at the end of the project
C Writing code without version control
D Deploying to production once a year

Part C · Answer key

Q	Ans	Correct option	Why
C1	B	Network (Layer 3)	Routers and IP work at Layer 3.

Q	Ans	Correct option	Why
C2	**B**	Data Link (Layer 2)	Switches forward frames using MAC addresses (Layer 2). Hubs are Layer 1 and routers are Layer 3.
C3	**C**	443	HTTP uses port 80 and HTTPS uses port 443.
C4	**D**	DHCP	DHCP follows DORA: Discover, Offer, Request, Acknowledge.
C5	**A**	192.168.1.10	The private ranges are 10.0.0.0/8, 172.16.0.0–172.31.255.255 and 192.168.0.0/16. 172.32.x.x is outside the range.
C6	**B**	62	That leaves 32 - 26 = 6 host bits, and 2^6 - 2 = 62. Two addresses are reserved for the network and broadcast.
C7	**A**	TCP is connection-oriented and reliable; UDP is connectionless and faster	TCP sets up a connection with a 3-way handshake and guarantees delivery. UDP has no handshake and is used for streaming and DNS.
C8	**B**	Not Found	2xx means success, 3xx redirect, 4xx client error (404 = not found, 403 = forbidden) and 5xx server error (500).
C9	**C**	AES	AES uses one shared key. RSA is asymmetric. SHA-256 is hashing, not encryption.
C10	**D**	Ransomware	Ransomware holds data hostage. A worm self-replicates across networks without a host file.
C11	**C**	SQL Injection	Malicious SQL in the input changes the query. Prevent it with parameterised queries.
C12	**B**	Confidentiality, Integrity, Availability	These are the three core goals of information security.
C13	**D**	IaaS	IaaS gives you infrastructure: VMs, storage and networks. You manage the OS and everything above it.
C14	**B**	SaaS	SaaS is ready-to-use software. The provider manages everything.
C15	**B**	Hybrid cloud	Hybrid combines private and public clouds.
C16	**B**	git clone	git init creates a new empty repo. git pull updates an existing one.
C17	**D**	A running instance of an image	Image = template, container = running instance, Dockerfile = build recipe. Containers share the host kernel.
C18	**A**	Merging code often, with automated build and tests on each change	CD (Continuous Delivery/Deployment) then automates releasing the tested code.

PART D · ≈ 14 QUESTIONS

CS Fundamentals (Programming)

D

The basics every programmer must know: data types, operators, control flow, functions, data structures, searching and sorting, Big-O and DBMS. Expect a mix of concept and small-logic questions.

Data types | Control flow | Stack · Queue · Linked list | Search & sort | Big-O | DBMS & SQL

D1

THE BUILDING BLOCKS

Data Types, Operators & Control Flow

Java type	Size	Range / note
byte	1 B	−128 to 127
short	2 B	−32,768 to 32,767
int	4 B	about ±2.1 billion
long	8 B	very large integers
float / double	4 B / 8 B	decimals (double is more precise)
char	2 B	Unicode (C's char = 1 byte)
boolean	—	true / false

Type casting

Implicit (widening): int → long → float → double, done automatically.
Explicit (narrowing): `(int) 9.7` = 9, which truncates.

Integer overflow

Going past the maximum wraps around: in Java, `Integer.MAX_VALUE + 1` becomes the most negative int.

Integer vs real division

`7 / 2 = 3` but `7.0 / 2 = 3.5`. `-7 / 2 = -3` in C/Java (truncates toward 0).

if / else-if / switch

In a switch, a **missing break** falls through to the next case.

for / while / do-while

do-while always runs **at least once**, because the condition is checked at the end.

Short-circuit

In `a && b`, b isn't evaluated if a is false. In `a || b`, b isn't evaluated if a is true.

Functions: call by value vs reference

By value: a copy is passed, so the original is unchanged. **By reference**: the function works on the original. Java passes everything by value (for objects, the value is the reference).

Arrays & strings

An array is a fixed-size, contiguous block, 0-indexed, with O(1) access by index. Java Strings are **immutable**. Use StringBuilder for repeated changes. A C string ends with `'\0'`.

D2

ORGANISE DATA

Data Structures

Stack: LIFO

`push`, `pop`, `peek`, all O(1). Overflow when full, underflow when empty.
Uses: function calls (the call stack), undo, browser back button, balanced brackets, expression evaluation, DFS.

Queue: FIFO

`enqueue` at the rear, `dequeue` from the front, O(1).
Variants: circular queue, deque (both ends), priority queue (by priority, often a heap).
Uses: CPU scheduling, printers, BFS.

Linked list

Nodes of `data + next` pointer. Insert or delete at the head is **O(1)**. Access by index is **O(n)**. Types: singly, doubly (prev + next) and circular.

Trees & graphs

Binary Search Tree: left < node < right, with search O(log n) on average and O(n) when skewed. Traversals: in-order (sorted output), pre-order, post-order. A **graph** has vertices and edges. BFS uses a queue; DFS uses a stack or recursion.

Structure	Access	Search	Insert	Delete
Array	O(1)	O(n)	O(n)	O(n)
Linked list	O(n)	O(n)	O(1) at head	O(1) at head
Stack / Queue	—	O(n)	O(1)	O(1)
BST (balanced)	—	O(log n)	O(log n)	O(log n)
Hash table	—	O(1) average	O(1) average	O(1) average

Expression notation (stack questions)

Infix	Prefix (Polish)	Postfix (Reverse Polish)
`A + B`	`+ A B`	`A B +`
`A + B * C`	`+ A * B C`	`A B C * +`
`(A + B) * C`	`* + A B C`	`A B + C *`

Evaluating postfix: scan left to right. Push numbers. When you meet an operator, pop the top two, apply the operator as (second-popped op first-popped), and push the result. Example: `5 1 2 + 4 * + 3 -` → 1 + 2 = 3, 3 × 4 = 12, 5 + 12 = 17, 17 − 3 = **14**.

D3

FIND & ORDER

Searching, Sorting & Big-O

Linear search

Checks each element: O(n). Works on unsorted data.

Binary search

Needs **sorted** data. Compare with the middle and discard half each step: O(log n). Searching 1,024 items takes at most 10–11 comparisons.

Algorithm	Best	Average	Worst	Space	Stable?	Idea
Bubble	O(n)*	$O(n^2)$	$O(n^2)$	O(1)	Yes	Swap adjacent pairs; the largest "bubbles" to the end
Selection	$O(n^2)$	$O(n^2)$	$O(n^2)$	O(1)	No	Pick the minimum and put it in front
Insertion	O(n)	$O(n^2)$	$O(n^2)$	O(1)	Yes	Insert each item into the sorted left part
Merge	O(n log n)	O(n log n)	O(n log n)	O(n)	Yes	Divide, sort halves, merge
Quick	O(n log n)	O(n log n)	**$O(n^2)$**	O(log n)	No	Partition around a pivot
Heap	O(n log n)	O(n log n)	O(n log n)	O(1)	No	Build a max-heap and extract repeatedly

*With the "stop if no swaps" optimisation, on already-sorted input.

Big-O from code

Pattern	Complexity
Fixed number of steps	O(1)
Halving or doubling loop (`i = i*2` , binary search)	O(log n)
Single loop to n	O(n)
Loop to n containing a halving loop; merge sort	O(n log n)
Two nested loops to n	$O(n^2)$
Naive recursive Fibonacci; all subsets	$O(2^n)$

Rules: drop constants (O(2n) = O(n)), keep only the dominant term ($O(n^2 + n) = O(n^2)$). Sequential blocks add, and nested blocks multiply. Growth order: $1 < \log n < n < n \log n < n^2 < 2^n < n!$

STORE DATA

D4 DBMS Basics

DBMS vs RDBMS

An **RDBMS** stores data in related **tables** (rows = records/tuples, columns = attributes) and uses SQL. Examples: MySQL, Oracle, PostgreSQL, SQL Server. NoSQL examples: MongoDB, Cassandra.

ACID (transactions)

Atomicity: all or nothing. **C**onsistency: valid state to valid state. **I**solation: concurrent transactions don't interfere. **D**urability: committed data survives crashes.

Key	Definition
Super key	Any set of columns that uniquely identifies a row
Candidate key	A minimal super key (no extra columns)
Primary key	The chosen candidate key: **unique + NOT NULL**, one per table
Alternate key	Candidate keys not chosen as primary
Foreign key	A column referencing another table's primary key (referential integrity)
Composite key	A key made of 2 or more columns

Key	Definition
Unique key	Values must be unique but can be NULL (one NULL in most databases)

Normal form	Rule
1NF	Atomic (single) values, no repeating groups
2NF	1NF + no **partial** dependency on part of a composite key
3NF	2NF + no **transitive** dependency (non-key → non-key)
BCNF	Every determinant is a candidate key

SQL family	Commands
DDL (structure)	CREATE, ALTER, DROP, TRUNCATE, RENAME
DML (data)	INSERT, UPDATE, DELETE
DQL (query)	SELECT
DCL (access)	GRANT, REVOKE
TCL (transactions)	COMMIT, ROLLBACK, SAVEPOINT

DELETE
DML. Removes rows (can use WHERE). Can be rolled back. Slower.

TRUNCATE
DDL. Removes **all** rows quickly but keeps the table structure.

DROP
DDL. Removes the whole table: structure and data.

Query order and joins. Written order: `SELECT … FROM … WHERE … GROUP BY … HAVING … ORDER BY`. WHERE filters rows **before** grouping, and HAVING filters **groups**. INNER JOIN returns only matching rows. LEFT JOIN returns all left rows (NULL where there's no match), and RIGHT JOIN returns all right rows. An ER diagram shows **entities** (rectangles), **attributes** (ovals) and **relationships** (diamonds) with cardinality 1:1, 1:N or M:N.

TEST YOURSELF

Part D · Practice MCQs

A mix of concept and small-logic questions. Aim for under 45 seconds each.

QD1 Size of an `int` in Java:

A 2 bytes
B 4 bytes
C Depends on the OS
D 8 bytes

QD2 Which data structure works on LIFO (Last In, First Out)?

A Stack
B Array
C Queue
D Linked list

QD3 Starting with an empty stack: push 1, push 2, pop, push 3, pop, pop. In what order are values popped?

A 1, 2, 3
B 3, 2, 1
C 2, 3, 1
D 2, 1, 3

QD4 Evaluate the postfix expression `2 3 4 * +`:

A 9
B 24
C 14
D 20

QD5 Binary search requires that the array is:

A Of even length
B Stored as a linked list
C Unsorted
D Sorted

QD6 Time complexity of this code?

```
for (i = 1; i < n; i = i * 2)
    print(i)
```

A O(log n)
B O(n)
C O(n log n)
D O(1)

QD7 Time complexity of two nested loops that each run n times:

A $O(n^2)$
B O(log n)
C O(2n)
D O(n)

QD8 Which sorting algorithm has a worst case of O(n log n)?

A Insertion sort
B Merge sort
C Quick sort
D Bubble sort

QD9 Which sorting algorithm is NOT stable (in its standard form)?

A Insertion sort
B Selection sort
C Merge sort
D Bubble sort

QD10 **Inserting a new node at the HEAD of a singly linked list takes:**

A O(1)
B O(n)
C O(log n)
D $O(n^2)$

QD11 **A PRIMARY KEY column must be:**

A A foreign key
B Unique and NOT NULL
C Repeated in every row
D Unique but may be NULL

QD12 **Which command is DDL (Data Definition Language)?**

A CREATE
B SELECT
C UPDATE
D INSERT

QD13 **Which statement about DELETE and TRUNCATE is correct?**

A TRUNCATE can use WHERE
B DELETE removes the table structure
C DELETE can use WHERE; TRUNCATE removes all rows
D They are both DDL

QD14 **A table in 2NF that still has a transitive dependency (non-key → non-key) violates:**

A 2NF
B 3NF
C 1NF
D No normal form

QD15 **'A transaction happens completely or not at all' describes which ACID property?**

A Durability
B Consistency
C Atomicity
D Isolation

QD16 **Which data structure is used for Breadth-First Search (BFS)?**

A Hash table
B Stack
C Heap
D Queue

Part D · Answer key

Q	Ans	Correct option	Why
D1	B	4 bytes	Java's sizes are fixed: byte 1, short 2, int 4, long 8, float 4, double 8, char 2.
D2	A	Stack	Stacks are used for undo, function calls and balancing brackets. Queues are FIFO.
D3	C	2, 3, 1	pop → 2, then 3 is pushed and popped → 3, then the last pop → 1.
D4	C	14	Push 2, 3, 4. Then * gives 3 × 4 = 12, and + gives 2 + 12 = 14.
D5	D	Sorted	Binary search halves a sorted range each step, so it runs in O(log n).
D6	A	O(log n)	i doubles each time, so the loop runs about $\log_2 n$ times.
D7	A	$O(n^2)$	n × n iterations.
D8	B	Merge sort	Quick sort's worst case is $O(n^2)$. Merge sort is always O(n log n).

Q	Ans	Correct option	Why
D9	**B**	Selection sort	A stable sort keeps equal elements in their original order. Selection sort's long-distance swaps can break that order.
D10	**A**	O(1)	Just point the new node at the old head. No traversal is needed.
D11	**B**	Unique and NOT NULL	A table has one primary key, which identifies each row. A foreign key links to another table's primary key.
D12	**A**	CREATE	DDL: CREATE, ALTER, DROP, TRUNCATE. DML: INSERT, UPDATE, DELETE. DQL: SELECT.
D13	**C**	DELETE can use WHERE; TRUNCATE removes all rows	DELETE is DML and row-by-row, and it can be rolled back. TRUNCATE is DDL and removes all rows quickly. DROP removes the table itself.
D14	**B**	3NF	1NF: atomic values. 2NF: no partial dependency. 3NF: no transitive dependency.
D15	**C**	Atomicity	ACID: Atomicity, Consistency, Isolation, Durability.
D16	**D**	Queue	BFS explores level by level using a queue. DFS uses a stack, or recursion.

GET TEST-READY

14-Day Plan & Final Cheat Sheet

Days	Focus	What to do (60–90 minutes a day)
1–2	Part A · Office	Practise every Excel function in A1 on a real sheet. Learn the error table and 20 shortcuts. Do the Part A MCQs.
3–4	Part B · Operators & bits	Learn the precedence table. Convert 30 numbers to binary. Drill & | ^ ~ << >> until each takes under 10 seconds.
5–6	Part B · Loops & recursion	Trace 15 snippets **without paper**. Use the loop-count formulas. Solve B1–B18 under time.
7	Part B · OOPs	Explain the 4 pillars aloud with examples. Learn overloading vs overriding. Do B19–B26.
8–9	Part C · Networking	Memorise the OSI table, ports, IP classes and private ranges. Practise subnet host counts.
10	Part C · Security & Cloud	Learn the encryption, malware and attack tables, IaaS/PaaS/SaaS, Git and Docker. Do the Part C MCQs.
11–12	Part D · CS Fundamentals	Learn the data structure and sorting tables, Big-O patterns, keys, normal forms and SQL families. Do the Part D MCQs.
13	Full mock	Take 45 mixed questions in 45 minutes. Note every wrong answer and why you got it wrong.
14	Revise	Read this cheat sheet and the trap boxes only. Sleep well.

One-page cheat sheet

A · Office

- `$A$1` fixed · `$A1` column fixed · `A$1` row fixed · `F4` toggles
- VLOOKUP: first column, look right, FALSE = exact, not found = #N/A
- COUNT = numbers · COUNTA = non-empty · COUNTIF(range, criteria) · SUMIF(range, criteria, sum_range)
- #DIV/0! · #N/A · #VALUE! · #REF! · #NAME? · #### (narrow column)
- Slide Master = all slides · Animation = objects · Transition = between slides

B · Pseudocode + OOPs

- `x<<n = x·2ⁿ` · `x>>n = x/2ⁿ` · `~x = −(x+1)` · `x^x = 0`
- `+` before `<<` · `==` before `&` · integer `/` truncates
- Nested loops `j = i to n` run n(n+1)/2 times · `i *= 2` runs log n times
- Recursion: expand down, add up on the way back. Printing after the call gives reverse order.
- Overloading = same class, different parameters · Overriding = child redefines, decided at run time

C · Networking · Security · Cloud

- L3 router/IP · L2 switch/MAC · L1 hub · L4 TCP/UDP
- 22 SSH · 25 SMTP · 53 DNS · 67/68 DHCP · 80 HTTP · 443 HTTPS
- Private: 10/8 · 172.16–31 · 192.168/16 · hosts = $2^h - 2$
- AES symmetric · RSA asymmetric · SHA hashing (one-way)
- IaaS = EC2 · PaaS = App Engine · SaaS = Gmail · Hybrid = private + public

D · CS Fundamentals

- Stack LIFO (DFS, undo) · Queue FIFO (BFS, scheduling)
- Binary search needs sorted data, O(log n) · Merge sort is always O(n log n) · Quick sort worst $O(n^2)$
- Stable sorts: bubble, insertion, merge · Not stable: selection, quick, heap
- PK = unique + not null · FK → another table's PK · 2NF partial · 3NF transitive
- DDL: CREATE/ALTER/DROP/TRUNCATE · DML: INSERT/UPDATE/DELETE · ACID

Final word

Parts A, C and D reward **revision**, and Part B rewards **practice**. Lock in the easy recall marks first, then win the pseudocode section by tracing calmly: write nothing down, track only what changes, and spot the pattern.

Disclaimer: This is an independent preparation guide for informational purposes only. It is not affiliated with, endorsed by, or an official document of Accenture. The section pattern is based on publicly shared preparation material and may change. All questions are original practice questions, not real test questions. Always check the latest details on the official Accenture careers portal.